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Optimising deep learning for smart building energy prediction: A particle swarm approach with real-world validation

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ABSTRACT

Accurate short-term load forecasting (STLF) in smart buildings for the operation and control of HVAC and building systems is particularly challenging due to the complexity of system dynamics. While hybrid deep learning (DL) models, such as CNN- and LSTM-based architectures, offer high predictive performance, they are resource-intensive. Simpler models like multilayer perceptrons (MLP) are computationally efficient but typically lack accuracy for STLF prediction. This study proposes a novel approach, PSO-MLP, which leverages particle swarm optimisation (PSO) to enhance MLP performance by tuning architectural and learning hyperparameters. A novel incremental technique is employed to iteratively build deep MLP from previous optimal solutions, an area not commonly explored. The method is tested on a real-world smart building—'Nanterre 3' (N3) at CESI Paris—and benchmarked against advanced, hybrid DL models (MHA-CNN-LSTM and LSTM-DWT-CRT) using public datasets (UCI IHEPC and AMPds), demonstrating its effectiveness and generalisability across diverse datasets. The PSO-MLP model achieved ~ 88% accuracy in predicting energy consumption 24-hour ahead in the N3 building, resulting in a 25.4% reduction in MAPE compared to the original MLP, with a runtime of 130 min. On public datasets, PSO-MLP outperformed both the original MLP (by 45.3%) and the MHA-CNN-LSTM (by 29.6%). It also consistently outperformed the LSTM-DWT-CRT model across all error metrics. Incorporating feature selection further improved accuracy by 5.49% on AMPds, making the proposed model a competitive, lightweight alternative for smart building energy forecasting with direct applications in building management systems and improving energy efficiency, and enabling edge computing and smart grid control.

1. Introduction

The energy supply sector is the largest contributor to greenhouse gas emissions, responsible for approximately 35% of total global emissions. Residential and commercial buildings are responsible for over half of all electricity consumed [1]. Urbanisation and population growth are driving increased energy demand on buildings, therefore, reducing energy consumption from buildings remains a key issue when tackling energy conservation and reducing emissions. Energy efficiency policies and technologies play a critical role in tackling energy conservation. Energy consumption prediction enables energy providers to regulate the supply and demand of energy and plays a vital role in enabling the control of smart grids [2] and microgrids [3], which incorporate renewable power generation and batteries to provide a reliable power supply at the lowest running cost [4]. Furthermore, it can also be used to highlight when buildings are consuming more energy than usual, triggering a review of heating, ventilation, and air conditioning (HVAC) control decisions and informing decision-making strategies for building managers and owners [5]. Demand

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Nomenclature

Acronym	Definition
Adam	Adaptive Moment Estimation
AI	Artificial Intelligence
AMPds	Almanac of Minutely Power Dataset
ANN	Artificial Neural Network
BMS	Building Management System
CNN	Convolutional Neural Network
CRT	Collaborative Representation Transform
DL	Deep Learning
DWT	Discrete Wavelet Transform
EA	Evolutionary Algorithm
ECNN	Enhanced Convolutional Neural Network
FS	Feature Selection
GAP	Global Active Power
GSOS	Grouped Sequential Optimisation Strategy
HL	Hidden Layer
HVAC	Heating, Ventilation, and Air Conditioning
IHEPC	Individual Household Electric Power Consumption
IoT	Internet of Things
IRBDNN	Iterative ResBlocks-Based Deep Neural Network
K-SVD	K-means-Singular Value Decomposition
LM	Levenberg–Marquardt
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MHA	Multi-Headed Attention
ML	Machine Learning
MLP	Multilayer Perceptron
MSE	Mean Squared Error
N3	Nanterre 3
NCEI	National Centers for Environmental Information
NEAT	NeuroEvolution of Augmenting Topologies
NEM	National Electric Market
PSO	Particle Swarm Optimisation
PSOGS	Particle Swarm Optimisation and Grid Search Approach
ReLU	Rectified Linear Unit
RF	Random Forest
RMSE	Root Mean Squared Error
RBiLSTM-AM	Residual Bidirectional LSTM with Attention Mechanism
STLF	Short-Term Load Forecasting
SVR	Support Vector Regression
TCN	Temporal Convolutional Network
UCI	University of California, Irvine
U_b/L_b	Upper bound/Lower bound
WHL	Whole House Load
WHO	Whale Optimisation Algorithm

side management also requires forecasting energy demands to prioritise energy use at times of high renewable power generation and to reduce or shed HVAC loads when the grid is under strain [6], helping to stabilise the grid and reduce the cost of running the building [7].

Approaches such as Short-Term Load Forecasting (STLF) allow prediction of building energy consumption for a few minutes to a week ahead and is an active area of research [3]. STLF can potentially improve the running costs of buildings when combined with building management systems (BMS) [8,9], by delaying the use of some electrically powered plant during peak times and participating in demand response schemes to help balance the power grid [10]. STLF is a difficult problem [11], particularly when considering individual residential units, due to the highly random and complex nature of the consumption profile. In contrast, predicting the STLF for a block of many residential units, a larger building, or a district, benefits from amalgamating all the individual unit load profiles, thus creating a smoother cumulative load profile [12].

However, implementing accurate STLF in real building environments presents significant operational challenges that must be addressed for effective deployment. HVAC systems, which constitute a major source of global operational CO_2 emissions due to their high energy demands, require sophisticated control strategies that can adapt to complex building environments whilst incorporating learning from operational experience [13]. Building automation and control systems (BACS) face increasing demands for sustainability, energy efficiency, and occupant comfort, requiring facility managers to possess specific technical knowledge for effective planning and operation [14]. Moreover, building integration within smart grids for demand response participation encounters substantial barriers, including limited interoperability between buildings and energy systems, regulatory framework gaps, and insufficient automation technologies to support coordinated demand management [15]. These operational complexities underscore the need for robust, deployable forecasting solutions that can address real-world implementation challenges.

Recent advances in building systems and smart technologies have enabled the capture and storage of large amounts of data on occupant behaviour and energy use. This data, gathered through embedded sensors and IoT devices, can be used with machine learning (ML) and deep learning (DL) techniques to predict future energy consumption by identifying patterns in historical data [16]. This task is typically framed as a data prediction problem, where a high-order nonlinear function is learned to approximate past trends and forecast future values. Advanced DL techniques have been widely explored for STLF due to their improved efficiency in learning complex nonlinear relationships compared to simpler, shallower DL approaches [17]. These are typically constructed by combining several models, such as a Convolutional Neural Network (CNN) combined with Long Short-Term Memory (LSTM). Simple DL models include deep multilayer perceptrons (MLP) comprising multiple hidden layers (HL). However, training DL models remains a challenging task, as it requires optimisation of the model architecture (i.e. the number of HLs and the number of neurons in each layer), the learning rate, batch size, and a variety of other hyperparameters [17]. Existing approaches generally fall into one of the following categories: (i) adopting a 'standard' model configuration [18,19]; (ii) manually testing a limited set of hyperparameter options through trial and error; or (iii) performing a manual or automated grid search across all possible combinations of hyperparameters. These methods require progressively greater time and computational resources, often necessitating compromises in the step size for each hyperparameter to make the grid search computationally feasible. However, evidence indicates that such techniques often fail to identify the most effective model configuration, as they rarely yield the best possible performance from the DL model [20].

Evolutionary optimisation techniques have demonstrated strong potential in enhancing grid search procedures by reducing the computational resources and time required [21,22]. They can identify near-optimal solutions without the need to exhaustively evaluate every possible parameter combination [23,24]. These techniques can be employed to optimise DL hyperparameters, improving prediction performance by avoiding the limitations of a fixed hyperparameter grid. Unlike conventional grid search, they enable a finer-resolution exploration of the search space, capturing solutions that would otherwise lie between the discrete points defined by grid-based methods.

Efforts in the field of STLF in buildings have focused on state-of-the-art DL techniques, which have high computational costs. However, few studies have attempted to produce scalable, lightweight alternatives with comparable prediction accuracy [25]. This paper aims to demonstrate how a Particle Swarm Optimiser (PSO)—an evolutionary optimisation technique—can be leveraged to tune the architectural and learning hyperparameters of a foundational MLP, enabling it to achieve optimal performance in comparison with more complex state-of-the-art DL techniques. To facilitate the exploration of deeper MLP architectures, a novel iterative approach is introduced, which progressively builds upon previously identified optimal configurations by increasing the network depth to enhance energy prediction accuracy. In light of these challenges, this study addresses the following research question: Can an incrementally deepened MLP, optimised via particle swarm optimisation, achieve STLF performance comparable to or exceeding that of advanced hybrid DL models in smart building applications? To answer this, the paper develops and validates a PSO-based optimisation framework for MLP depth and hyperparameters, evaluates its performance, and analyses its potential for integration into BMS and demand response strategies. The proposed method is applied in a real-world case study using a full-scale demonstrator: the 'Nanterre 3' (N3) smart building, located on the CESI campus in Paris [16]. In addition, its performance is evaluated against two advanced state-of-the-art DL models—namely, the CNN-LSTM with multi-headed attention (MHA-CNN-LSTM) [26] and the collaborative representation transform LSTM model (LSTM-DWT-CRT) [27]—using two public datasets: the UCI Individual Household Electric Power Consumption (UCI IHEPC) dataset [28] and the Almanac of Minutely Power Dataset (AMPds) [29].

The proposed PSO-MLP model demonstrates significant potential for addressing the real-time operational challenges identified above. ML approaches for building energy management face substantial deployment limitations, with only 23% of studies demonstrating real building implementation due to computational complexity and extensive retraining requirements [13]. In contrast, the lightweight computational characteristics of the PSO-MLP approach make it particularly suitable for integration with existing building automation systems, where accurate STLF can support automated HVAC scheduling, coordinate with renewable energy sources, and enable responsive demand management strategies [14]. As buildings seek to participate more effectively in smart grid demand response programmes, such forecasting capabilities become essential for demand optimisation and coordination, addressing key technological barriers that currently limit building participation in grid-level energy management [15].

2. Related work

2.1. STLF in buildings

STLF plays a crucial role in supporting operational decision-making within power systems, including dispatch scheduling, load balancing, and reliability assessment [30]. Accurate STLF contributes to the development of cost-effective, secure, and stable

operational plans by preventing both overestimation and underestimation of demand, thereby reducing energy waste and mitigating the risk of energy shortages [31]. STLF methods are commonly categorised into deterministic point forecasts and probabilistic forecasts [32]. These forecasting techniques are further classified into four methodological groups: Statistical Techniques, Artificial Intelligence (AI) Techniques, Knowledge-Based Expert Systems, and Hybrid Approaches [33]. Among these, contemporary research has predominantly focused on statistical and AI-based methods, particularly ML approaches, due to their ability to capture complex, nonlinear patterns in electricity consumption data and to adapt effectively to changing demand dynamics [34]. A standard framework for STLF includes data acquisition, data preprocessing, model training, and load forecasting [35]. In smart building and grid contexts, more elaborate structures involve six phases, such as data collection, feature selection (FS), data processing, model training and testing, and optimal model selection [36]. Commonly employed AI techniques include Artificial Neural Networks (ANN), Support Vector Regression (SVR), LSTM, CNN, Random Forests (RF), and XGBoost, as well as hybrid, advanced DL models such as CNN-BiGRU [37], LSTM-CNN [38], Temporal Convolutional Networks (TCN) [39], and Iterative Resblocks-Based DNN (IRBDNN) [40], each offering specific advantages and facing distinct limitations [41,42]. Numerous studies have evaluated these methods across various real-world applications, such as the National Electric market (NEM) in Australia and the National Centers for Environmental Information (NCEI) in the United States [39]. Classical statistical models, such as ARIMA and SARIMA, continue to show strong performance in structured, time-dependent applications [43,44].

2.2. Deep learning techniques

DL techniques are commonly applied to STLF due to their strength in learning nonlinear relationships between input features [45]. Bu and Cho [26] employed a MHA mechanism with a CNN and LSTM to model the UCI IHEPC dataset [28]. The attention mechanism was shown to enhance the model's ability to predict sudden spikes in power consumption, a known limitation in prior DL studies. The MHA-CNN-LSTM model achieved a 31.01% error reduction compared to a conventional CNN-LSTM. Kim and Cho [46] had earlier used a CNN-LSTM model on the same dataset and reported higher error rates than Bu and Cho [26], although still outperforming models from [47,48]. They emphasised the challenges of hyperparameter tuning in DL and identified genetic algorithms as a promising solution for automating the search process. Gonçalves et al. [49] further improved forecasting accuracy using a Variable Split Convolutional Attention LSTM model. However, they used hourly resolution data to predict the following day's energy usage, while studies such as Bu and Cho [26] used daily time step data. Averaging 24 hourly data points into a single daily value, as done in some studies, sacrifices input data granularity and may reduce predictive power [12]. Imani and Ghassemian [27] implemented Discrete Wavelet Transform (DWT) and Collaborative Representation Transform (CRT) with LSTM to forecast residential load using the AMPds [29]. The data was downsampled to 30-minute intervals to simulate smart meter readings, and the load was predicted one day ahead. The DWT helped to separate the high- and low-frequency components of the data, whereas the CWT helped to reduce noise in the data. Forecasts were evaluated for both the six sub-meters and the aggregated whole house load. Kong et al. [50] applied LSTM to the AMP2ds benchmark dataset. AMPds contains only one year of data, whereas AMP2ds includes two years, which significantly affects model generalisation. With only one year of data, the test set includes patterns unseen in training; with two years, temporal overlap (e.g. same season across years) improves performance. Accordingly, Kong et al. reported lower errors using only the whole house load data. Liu et al. [51] introduced a two-stage electricity load forecasting method combining deep sparse coding and edge computing, applied to the AMP2ds dataset. Their hybrid approach used a deep K-means-Singular Value Decomposition (K-SVD) model to extract the behaviour of residents and help improve the LSTM performance. Recent innovations have further enhanced the predictive power of STLF. For instance, XGBoost constructs collections of decision trees to iteratively reduce forecasting error [52]. The AFC-STLF model accelerates convergence while improving accuracy [53], and the Enhanced CNN (ECNN) outperforms traditional CNN models [54]. Moreover, the Residual Bidirectional LSTM with Attention Mechanism (RBiLSTM-AM) captures complex patterns in time-series data using residual connections and attention, significantly increasing forecast accuracy [55].

2.3. PSO and evolutionary optimisation

Combining Evolutionary Algorithms (EA) with DL techniques can yield superior results compared to using each method in isolation. However, there is considerable variation in how network information is encoded, and little consensus on the most effective approach [56]. Stanley and Miikulainen [57] introduced the NeuroEvolution of Augmenting Topologies (NEAT) technique, which simplifies neural network design through graph encoding. NEAT separates node and connection genes, enabling flexible recombination and mutation to enhance population fitness. Five years later, they developed HyperNEAT [58], which encodes genetic information indirectly via a compositional pattern-producing network, making it scalable for larger neural networks. Ly et al. [59] employed PSO to optimise weights and biases in an ANN for a regression problem. They evaluated 420 ANN architecture combinations, each with 1000 Monte Carlo simulations, splitting the data into training and testing sets for every run. The results revealed that larger models tend to perform better on training data, while more modest-sized models generalise better. Following the architecture search using the Levenberg–Marquardt (LM) algorithm, the authors re-trained the optimal ANN using PSO, which consistently outperformed LM-trained models. This highlights the importance of both ANN architecture selection and weight optimisation. However, while effective for simple regression problems, this method is computationally prohibitive for time-series forecasting [60]. Moradi et al. [61] applied two EA—PSO and the Whale Optimisation Algorithm (WHO)—to automatically tune the architecture of a MLP model with up to four HL. They observed significant performance gains for models with two HL compared to single-layer configurations. However, their optimisation was limited to architecture alone, excluding further hyperparameter tuning.

Amasyali and El_Ghorary [62] noted the limited application of DL in building energy consumption prediction. Using simulated energy consumption data, they assessed ANN with up to four HL and found that 4-HL models perform best when sufficient data is available, whereas single HL models are preferable when data is limited. Bacanin et al. [20] demonstrated that EA can successfully optimise four key hyperparameters of an LSTM model — (i) number of neurons, (ii) learning rate, (iii) training epochs, and (iv) dropout rate — for improved STLF. Their EA-tuned models outperformed those generated via grid search. Nevertheless, the study was restricted to a single LSTM layer and just five iterations per EA run, due to LSTM's computational demands. Acikkar and Altunkol [63] proposed a hybrid PSO and Grid Search approach (PSOGS) that improves efficiency over standalone Grid Search or PSO. In PSOGS, particle positions are constrained to a predefined grid, allowing the reuse of previously evaluated fitness values. When optimising three hyperparameters of a SVR model, PSOGS achieved competitive accuracy using fewer SVR runs. However, its scalability is questionable, as increasing the number of hyperparameters reduces the probability of revisiting grid positions. Furthermore, fixing MLP hyperparameters to enable grid positioning may compromise predictive accuracy compared to fully optimised PSO solutions.

2.4. Research gaps and contributions

As buildings and energy systems become more intelligent, data-centric and decentralised, robust STLF models remain critical to enabling flexible, responsive, and sustainable grid operation. The previous sections show that advanced, hybrid DL models offer superior accuracy through the ability to capture complex temporal patterns for STLF. However, they present significant implementation and operational challenges when compared to simple, foundational models like MLP [64]. The latter are relatively shallow, involve fewer trainable parameters, and are relatively easier to configure and train, requiring basic tuning of learning rate, batch size, and the number of HL [65]. In contrast, LSTM- or Transformer-based models involve millions of parameters, multiple architectural decisions (e.g. number of layers, attention heads, memory sizes), and extensive hyperparameter optimisation. For example, tuning dropout rates, sequence lengths, optimiser types (e.g. Adam vs. RMSprop), and learning rate schedules can dramatically affect performance and stability [64]. Moreover, while MLP can be trained on standard CPUs within minutes, advanced models generally require GPU acceleration due to their computational demands, especially when working with large-scale time-series data from regional or national power grids. Training times can range from several hours to multiple days, depending on data volume and model complexity. For example, Transformer-based models consume substantial memory due to their quadratic attention mechanisms, resulting in high computational and energy costs [66].

On the other hand, training simple DL models, such as MLP, becomes increasingly challenging as deeper architectures are considered, making it difficult to develop optimal models that achieve strong STLF performance. This explains why most existing studies employ only shallow MLP—typically with no more than four HL—and primarily use them as baseline models with relatively poor performance [67]. Most prior PSO-MLP works are limited to these shallow networks without feature selection, whilst our contribution provides an incremental deepening strategy with FS validated on real buildings.

This highlights a clear research gap: the need to investigate how foundational DL models, such as deep MLP, can be trained and optimised more effectively, both in terms of architectural design and learning configurations, to achieve STLF performance comparable to that of more advanced, hybrid DL techniques.

The main contributions of this paper are as follows:

1. The development of a novel incremental depth, iterative PSO-based technique for constructing optimal deep MLP models with integrated feature selection, addressing the limitations of existing shallow PSO-MLP approaches.
2. The demonstration of the proposed approach through a real-world case study involving a full-scale demonstrator—the N3 Smart Building, located on the CESI campus in Paris [16].
3. The evaluation of the solution performance on two additional public STLF datasets, with detailed results and analysis confirming that the proposed method outperforms advanced state-of-the-art DL models, such as MHA–CNN–LSTM [26] and LSTM–DWT–CRT [27].

3. Research methods and tools

This study implemented a structured five-step methodological framework to develop, optimise, and evaluate the PSO-MLP model for smart buildings STLF (Fig. 1). The process integrates data-driven modelling and prediction combined with metaheuristic optimisation and comparative analysis to ensure robustness and applicability across different datasets.

3.1. Data capture and storage

Raw energy consumption data were collected and stored in a structured database to enable scalable access and analysis, using the CESI N3 Smart Building as a real-world case study [16] (Fig. 1 - Step 1). Located on the CESI campus in Paris, the N3 building serves as a demonstrator for energy-efficient smart building technologies. Constructed in 2017 from 16 recycled modular shipping containers, the 220 m^2 facility is equipped with an extensive network of over 150 sensors, including motion detectors and photometric cells, dedicated to monitoring lighting and occupancy conditions. These sensors continuously collect data from each of nine control zones spread over two floors, which are stored within a dedicated MariaDB relational database to support intelligent energy management and predictive control systems [68]. The building's operational environment has been modelled to capture detailed information on physical systems (e.g. HVAC, lighting, insulation) and smart components (e.g. IoT sensors), enabling structured time-series data capture, storage, and simulation [16]. Full details of the CESI N3 Smart Building are provided in [16].

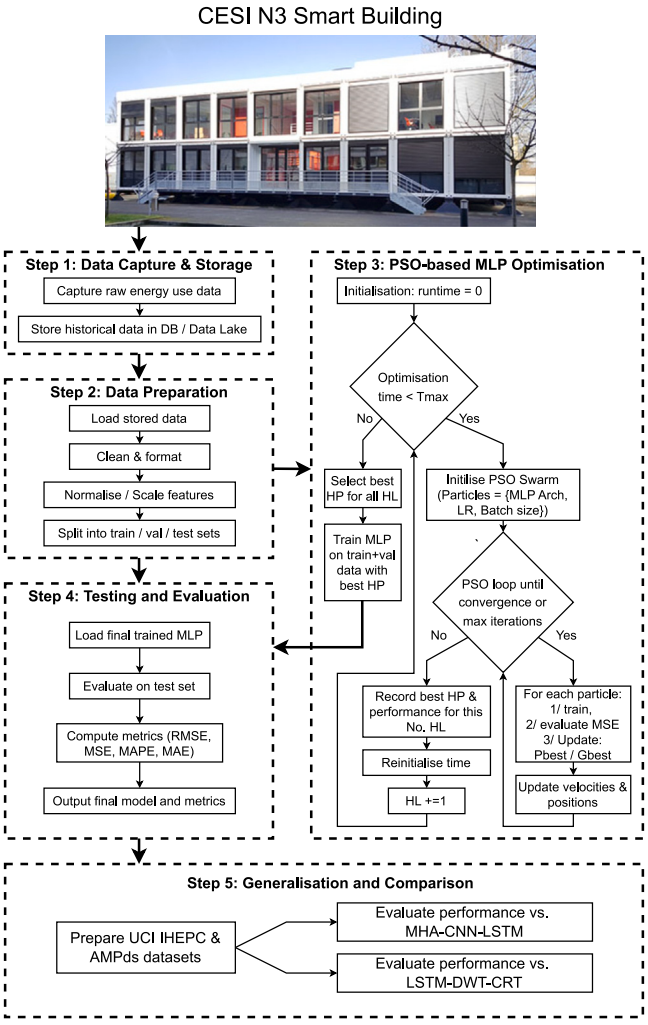


Fig. 1. Research methods and tools used to develop, optimise, and evaluate the PSO-MLP solution.

3.2. Data preparation

The stored data was cleaned, formatted, and normalised, and subsequently split 70:20:10 between training, validation and testing subsets to support model development (Fig. 1 - Step 2). The N3 building dataset comprises 523,553 rows of data, recorded at one-minute intervals over a one-year period from 1 June 2023 to 31 May 2024 [68]. During data preparation, 2047 rows (0.39%) were identified as missing; these were imputed using values from the same time of day two weeks earlier. No outlier detection was carried out, and care was taken to ensure that time stamps for each data series aligned correctly. The temporal resolution was then adjusted from one-minute to half-hourly intervals. The building's total active power was selected as the dependent variable, with the aim of predicting it one day in advance. In addition to the timestamp, six input features were selected: total active power, the active power of phases A, B, and C, external light intensity, and external ambient air temperature. To incorporate temporal dependencies, lagged features were generated using values from the previous three hours and from the same three-hour window on the two preceding days. Data was standardised using Scikit-learn's Standard Scaler or Min_Max Scaler before being processed as inputs and ground truth for the PSO-MLP at the required time resolution. At the end of the process, the standardised PSO-MLP prediction was converted back to the original scale using the corresponding column statistics calculated prior to standardisation.

3.3. PSO-based MLP optimisation

A foundational MLP model was optimised using PSO, where three key hyperparameters (i) the number of neurons per HL; (ii) learning rate; and (iii) batch size, were explored within a time-constrained search space (Fig. 1 - Step 3). The optimisation process was designed not only to tune these parameters but also to iteratively explore deeper network architectures by incrementally

increasing the number of HL. The optimisation begins with a shallow network configuration (i.e. a single HL), and with each run of the PSO, the number of HL is incremented by one. For each HL configuration, PSO is used to find the best combination of the three hyperparameters. Once the best-performing MLP for the current HL is identified, the process continues by adding an additional HL, allowing the model to build upon the previously optimal structure and progressively explore deeper architectures. Within the PSO algorithm, each particle represents a candidate MLP solution encoding the three hyperparameters. The particle's position in the d-dimensional space ($d = 3$) is normalised to the range $[0, 1]$ within the PSO to ensure all dimensions contribute equally despite having different magnitudes. For example, learning rate values typically range from 0.005 to 0.02, whereas batch sizes may range between 30 and 450. The hyperparameter values are then extracted, unnormalised and rounded for number of neurons and batch size, and then passed to initialise and train a corresponding MLP. At each run of the PSO, a separate MLP is trained for each particle's configuration. These models are trained using the Mean Squared Error (MSE) as the loss function, with traditional early stopping applied to prevent overfitting; training is halted if no improvement is observed in the validation set for three consecutive epochs [69] (with a maximum of 100 epochs). The trained model is evaluated on the validation dataset, and the resulting MSE is used as the particle's fitness score within the swarm. After completing all PSO iterations for a given HL configuration, the best-performing model is recorded and the process moves to the next HL depth, incrementally increasing complexity. This loop continues until the total optimisation time reaches the predefined threshold ($Tmax$) of 3 h. Once the best configuration across all HL levels has been identified, the final MLP is retrained on the combined training and validation data using the optimal hyperparameters and results are generated using the test data (Fig. 1 - Step 4). The parameters used in the PSO are explained in Section 4.1.3.

3.4. Testing and evaluation

The optimised MLP was evaluated on an independent test set from the N3 Smart Building dataset. Performance metrics, including Root Mean Squared Error (RMSE) [70], Mean Absolute Percentage Error (MAPE) [71], and Mean Absolute Error (MAE) [72], were computed to assess predictive accuracy (Fig. 1 - Step 4). These metrics are directly aligned with building engineering objectives: accurate short-term load forecasts support both monitoring of energy consumption trends and control of building systems, such as optimising HVAC operation and enabling demand response strategies. By evaluating prediction errors, building managers can quantify the reliability of the forecasts for operational decision-making in smart building environments.

3.5. Generalisation and comparison

The proposed solution was further tested and validated on two benchmark datasets—UCI IHEPC [28] and AMPds [29]—to assess its generalisability. These two domestic building electrical power datasets were chosen to test our PSO-MLP with highly random and complex consumption profiles, which are more challenging to model than those for larger buildings or groups of buildings [12]. The PSO-MLP performance was compared against two advanced state-of-the-art DL architectures: MHA-CNN-LSTM [26] and LSTM-DWT-CRT [27] (Fig. 1 - Step 5). These evaluations also demonstrate the relevance of the proposed model for building-level energy monitoring and management. The same data preprocessing described above for the CESI dataset was also applied to these two benchmark datasets. The UCI IHEPC dataset was processed at multiple temporal resolutions to perform one-step-ahead load forecasting, enabling a direct comparison with the approach in [26]. Global Active Power (GAP) was selected as the dependent variable. All available input features, with the exception of Global Reactive Power, were used to generate lagged input variables across various time steps, depending on the temporal resolution. The lagged features reflect practical building operational cycles, capturing daily and weekly usage patterns. For example, at a one-hour resolution, in addition to six calendar-based time features (e.g. hour, day, month), seven input features were extracted from the current time step and the three preceding hours, as well as from the same four time steps on the previous day, two days earlier, and one week earlier, yielding a total of 117 input features. In contrast, for coarser resolutions ranging from one to seven days, only the previous seven time steps were used in combination with the six calendar features, resulting in 55 input features. For the AMPds dataset, and to enable comparison with the work in [27], two predictive experiments were conducted, both targeting one-day-ahead load forecasting. In the first, six appliance loads were selected from the 21 available: Clothes Washer, Clothes Dryer, Wall Oven, Heat Pump, Dishwasher, and TV. As these loads account for only a small fraction of total household energy consumption (and frequently report zero values throughout the day), their combined usage was subtracted from the Whole House Load (WHL) to construct a more informative input feature set. This ensures the model captures appliance-level patterns relevant for building energy management. The second experiment used WHL as unique input, reflecting common practice in smart-metring applications for domestic energy monitoring [73]. In both experiments, lagged variables were engineered to align with the methodology in [27], incorporating WHL values from the previous three hours, the same three-hour window from the previous day and week, and the current WHL value from those same historical periods. This feature engineering ensures alignment with operational cycles and supports evaluation metrics meaningful for building monitoring and control.

4. Experimental setup and evaluation

4.1. Experimental setup

4.1.1. Computing environment

The PSO-MLP model was implemented in Python 3.10 using PyTorch 2.0.1 and executed on the Google Colaboratory Pro+ cloud-based platform. A Tesla T4 GPU with 16 GB of VRAM was used for model training and evaluation, except for the one-minute resolution UCI IHEPC dataset, which required a high-RAM session with 51 GB of system memory.

Table 1
CESI N3 Smart Building results PSO-MLP vs Standard MLP.

	PSO-MLP (FS)	Standard MLP
MAPE	12.190	16.351
MAE	426.13	546.84
MSE	316704	453288
Runtime	2 h 10 m 20 s	17.86 s
Architecture	32,99,117,110,146,69,82,48,150,26,1	65
Batch Size	275	100
Learning Rate	0.0005703726783394814	0.001

4.1.2. MLP configuration

The Rectified Linear Unit (ReLU) activation function was used in all HL of the MLP model, while a linear activation was applied in the output layer to produce the required continuous regression output. ReLU is a suitable choice for training deeper MLP architectures due to its computational efficiency and ability to mitigate vanishing gradient issues [74]. Each MLP initialised from a PSO particle is assigned unique random weights and biases to preserve diversity within the swarm, while the MLP model's hyperparameters are defined by the particle's multi-dimensional position in the search space. To perform backpropagation, the Adaptive Moment Estimation (Adam) optimiser available in PyTorch was employed, owing to its demonstrated effectiveness in training deep neural networks.

To ensure repeatability and to enable the PSO to search effectively, it was important to be able to evaluate and report on one result per particle at each iteration. To achieve this, the random seed value was set to 42 at the start of every run, as carried out in [75], and the random initialisation of weights and bias values for each particle's MLP for the first iteration was used for all subsequent PSO iterations. This ensured that each particle's MLP started its training from the same state each time.

4.1.3. PSO-based MLP optimisation process

The PSO cognitive and social constants were set to 1.75. The inertia weight was initially set to 0.7 and linearly decreased to a minimum of 0.2 by the final PSO iteration. Particle velocity and position updates were constrained to ensure all particles remained within the defined search space bounds. A swarm size of 15 particles was used throughout, as this configuration was found to offer a suitable balance between convergence performance and computational efficiency. A maximum of 100 PSO iterations was permitted for each experiment, except for the UCI IHEPC dataset at one-minute resolution, where the number of iterations was limited to 25 due to increased runtime requirements. A PSO dimension convergence plot was generated for each run to monitor optimisation progress, verify that the upper and lower bounds for each dimension were appropriate, and confirm that convergence occurred within these bounds.

The range of each hyperparameter in the PSO search space is defined by upper (U_b) and lower (L_b) bounds, which allow for control over how the optimal multi-HL architecture is explored. Rather than performing a broad search across multiple depths simultaneously, this study adopts an incremental strategy: the optimal configuration is first identified for a single HL, and this result is then used to inform the search for the optimal configuration with two HL, and so on. For the initial one-HL run, the values of U_b and L_b for each PSO dimension were determined through a combination of prior experience and trial-and-error. If the PSO convergence plot indicated that a solution was constrained by one or more boundary limits, the result was discarded and the corresponding bounds were adjusted before rerunning the optimisation. The optimal solution obtained from the one-HL configuration was then used to guide the adjustment of U_b and L_b for the subsequent two-HL search. This iterative process was repeated as the MLP architecture was progressively deepened. From the range of solutions produced, the highest performing MLP architecture with its optimal learning hyperparameters is used to test the PSO-MLP performance with the test dataset (Fig. 1 - Step 4).

4.2. Results and analysis

4.2.1. CESI N3 smart building case study

The model effectively predicted the energy pattern 24 h in advance at a 30-minute resolution. FS using a RF [76] was employed to identify the 32 most significant inputs. Although the feature selection used is a conventional method and usually lacks exploration of new feature engineering [77], it enabled an average improvement across all three error metrics of 6% resulting in a test data MAPE of 12.2%. Table 1 presents the results of the PSO-MLP with FS alongside those of an MLP configured with standard HP, used here as a baseline. The standard MLP architecture was defined following the guidance in [19]. Fig. 2 shows the predicted vs. actual values for the week from Thursday, 23 May to Wednesday, 29 May 2024. The PSO-MLP model tuning resulted in a 25.4%, 22.1% and 30.1% reduction in MAPE, MAE and MSE, compared to the baseline MLP model with manually set hyperparameters. The trained MLP model running on an Intel Xeon CPU with a clock speed of 2.20 GHz has a single time step inference time of less than 3 ms.

4.2.2. Evaluation on public benchmark datasets

The MSE, MAE and MAPE of the prediction on the test data are reported to allow comparison with two advanced state-of-the-art DL architectures: MHA-CNN-LSTM [26] and LSTM-DWT-CRT [27]. The optimal model architecture, including input and output layers, is provided in the results tables in the column headed 'PSO-MLP (Architecture)'. The PSO-MLP is used to search for optimal MLP models for both datasets.

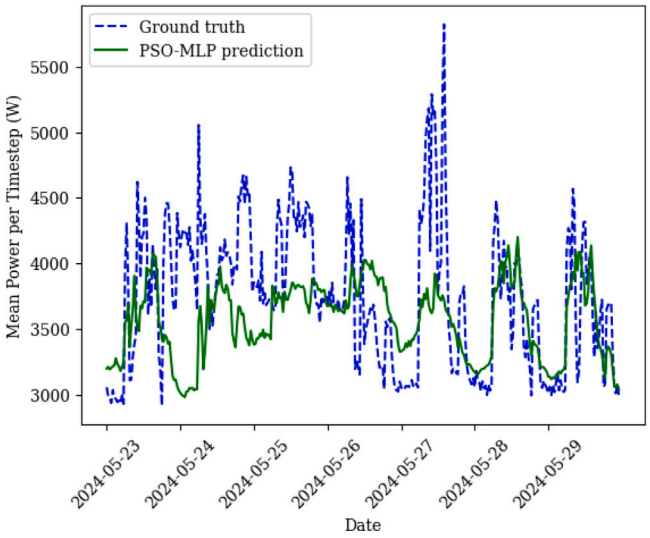


Fig. 2. Actual vs Predicted Energy Use (CESI N3 Smart Building).

Table 2
UCI IHEPC dataset: comparison of MSE at different time step resolutions with MHA-CNN-LSTM model.

Resolution	MLP [26]	MHA-CNN -LSTM [26]	PSO-MLP (MSE)	PSO-MLP (Architecture)	Learning rate	Batch size
1M	0.1003	0.0516	0.0431	81:68:67:73:61:66:-81:56:63:92:43:13:5:1	0.00021	579
15M	0.2370	0.1838	0.1586	81:77:76:66:65:60:-48:52:28:6:1	0.00608	351
30M	0.2813	0.2366	0.2027	81:178:153:147:104:-112:118:149:154:168:-171:129:77:42:16:1	0.00692	131
45M	0.3208	0.2838	0.2338	81:10:12:5:1	0.00322	261
1H	0.3072	0.2662	0.2047	117:117:104:94:60:-46:36:23:16:1	0.00100	125
1D	0.1134	0.0969	0.0445	57:114:96:89:71:-49:8:1	0.01245	272
7D	0.0441	0.0305	0.0098	57:58:51:60:56:57:-55:36:36:26:8:1	0.00925	42

UCI IHEPC dataset. Table 2 shows the results for the PSO-MLP model using data standardised with the 'Min_Max' Scaler. It presents the MSE results and MLP hyperparameters for the UCI IHEPC dataset using the proposed PSO-MLP model across seven different time resolutions (shown on the right), alongside two comparative sets of MSE values: one from a baseline MLP and the other from the MHA-CNN-LSTM model [26], shown on the left. The PSO-MLP architecture specifies the number of neurons in each layer. The prediction at all time resolutions is for the next time step ahead.

The MSE results shown in Fig. 3 reveal that PSO-MLP achieves better results than any of the comparator techniques (i.e. foundational MLP and MHA-CNN-LSTM) in all time resolutions.

Fig. 4(a) displays the UCI IHEPC test data at 30-minute resolution, labelled 'Ground truth' between 7th and 13th July 2010. The 'PSO-MLP prediction' is the 'global best' run from the PSO-MLP using the Standard Scaler. While the prediction tracks the ground truth quite effectively, there is some lag, and the PSO-MLP performs poorly at predicting rapid large changes in energy use. The prediction tends to overestimate the peak values and underestimate the lowest ones, but it follows the ground truth load profile. Fig. 4(b) shows the absolute error for the same time period as the PSO-MLP prediction shown in Fig. 4(a). The highest absolute errors do occur at times when the load changes rapidly, confirming that the PSO-MLP is not predicting these rapid changes in load effectively. However, outside of these time periods of rapid load change, the error rate is significantly lower. Fig. 5 shows the performance of both standardisation methods across all seven time resolutions. The best results were achieved using between three and fourteen HL, with a significant improvement typically observed when increasing from a single HL to deeper architectures.

AMPds dataset. Table 3 presents the results for the first AMPds study, which involves six appliance-level measurements, with and without FS for the PSO-MLP model (shown on the right-hand side). For comparison, two advanced DL models — FFN with CRT and

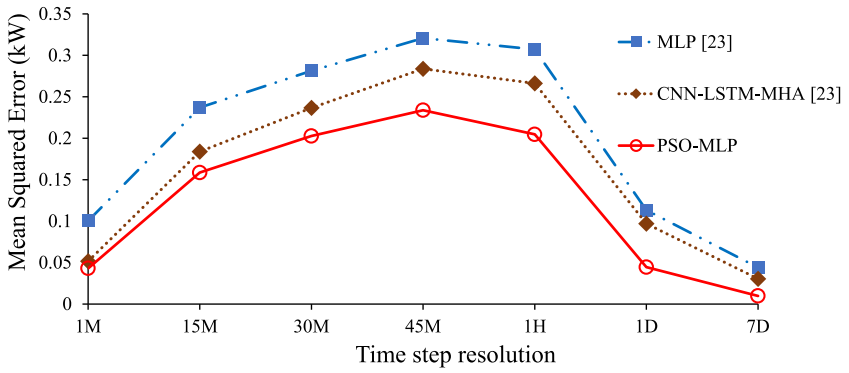


Fig. 3. UCI IHEPC Results.

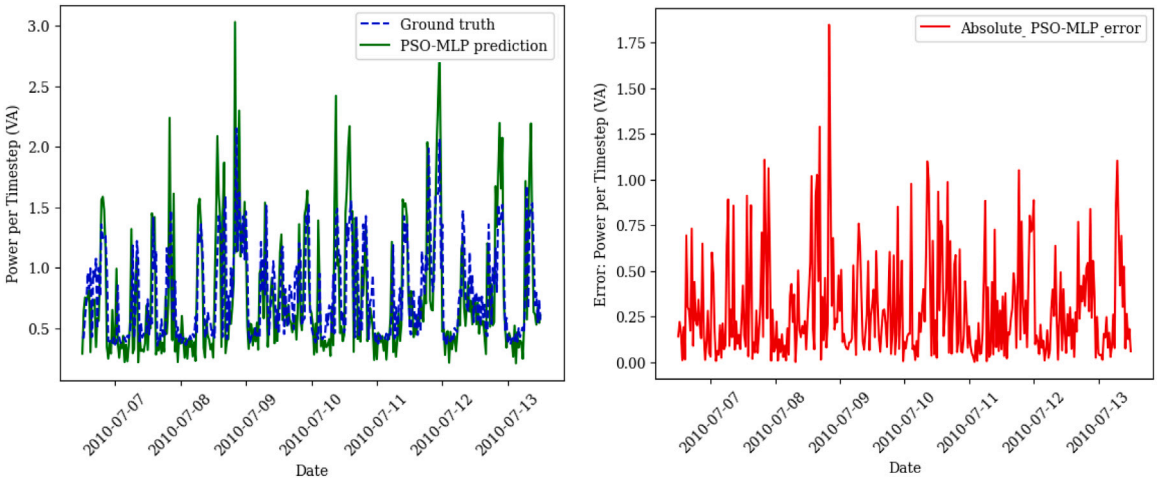


Fig. 4. UCI IHEPC 30 Minute resolution, Standard Scaler. PSO-MLP prediction vs Ground Truth and Absolute error.

Table 3
AMPds - six appliance measurements: Results comparison with LSTM-DWT-CRT [27].

Error metric	FNN with CRT [27]	LSTM-DWT-CRT [27])	PSO-MLP	PSO-MLP FS
MAPE	45.30	47.11	35.66	34.44
MAE	279.76	272.56	262.27	256.80
MSE	168904	155975	153392	149496

LSTM-DWT-CRT [27] — are included on the left-hand side. The 'FFN with CRT' is a MLP enhanced with a CRT, and serves as a baseline comparator for the PSO-MLP. Predictions are made one day ahead, equivalent to 24 h or 48 time steps, given the 30-minute time resolution. Data standardisation using the Min_Max Scaler produced the best results, as shown in Tables 3 and 4, and Fig. 6.

The PSO-MLP shows improved performance across all error metrics compared to both model results. The optimal MLP architecture in Table 3 was 155:122:100:78:58:53:40:27:25:22:13:1 (ten HL), with a learning rate of 0.00661 and a batch size of 242. FS was employed with this dataset based on the predicted importance of each input feature, as estimated by the permutation of out-of-bag predictor observations from a RF of 100 regression trees. This number of trees is commonly selected for regression problems of this type [76]. The RF technique is selected as it returns an 'importance value' for each feature, indicating the degree of usefulness in making the prediction. The 'importance value' is used to select features to retain within the dataset by setting a 'threshold value'. FS with a 'threshold value' of 0.005 selects 61 from the original 155 features, and the resultant PSO-MLP FS runs deliver improvements in all three error metrics. The optimal MLP architecture with FS is 61:151:139:131:128:117:101:93:82:67:75:63:39:24:13:1 (fourteen HL), with a learning rate of 0.00266 and a batch size of 242. The PSO-MLP results for six appliance loads following FS are better than the advanced state-of-the-art LSTM-DWT-CRT model. Table 4 presents the results using WHL. Again, all three error metrics for the PSO-MLP are better than those of the FNN with DWT, and also outperform the advanced state-of-the-art LSTM-DWT-CRT

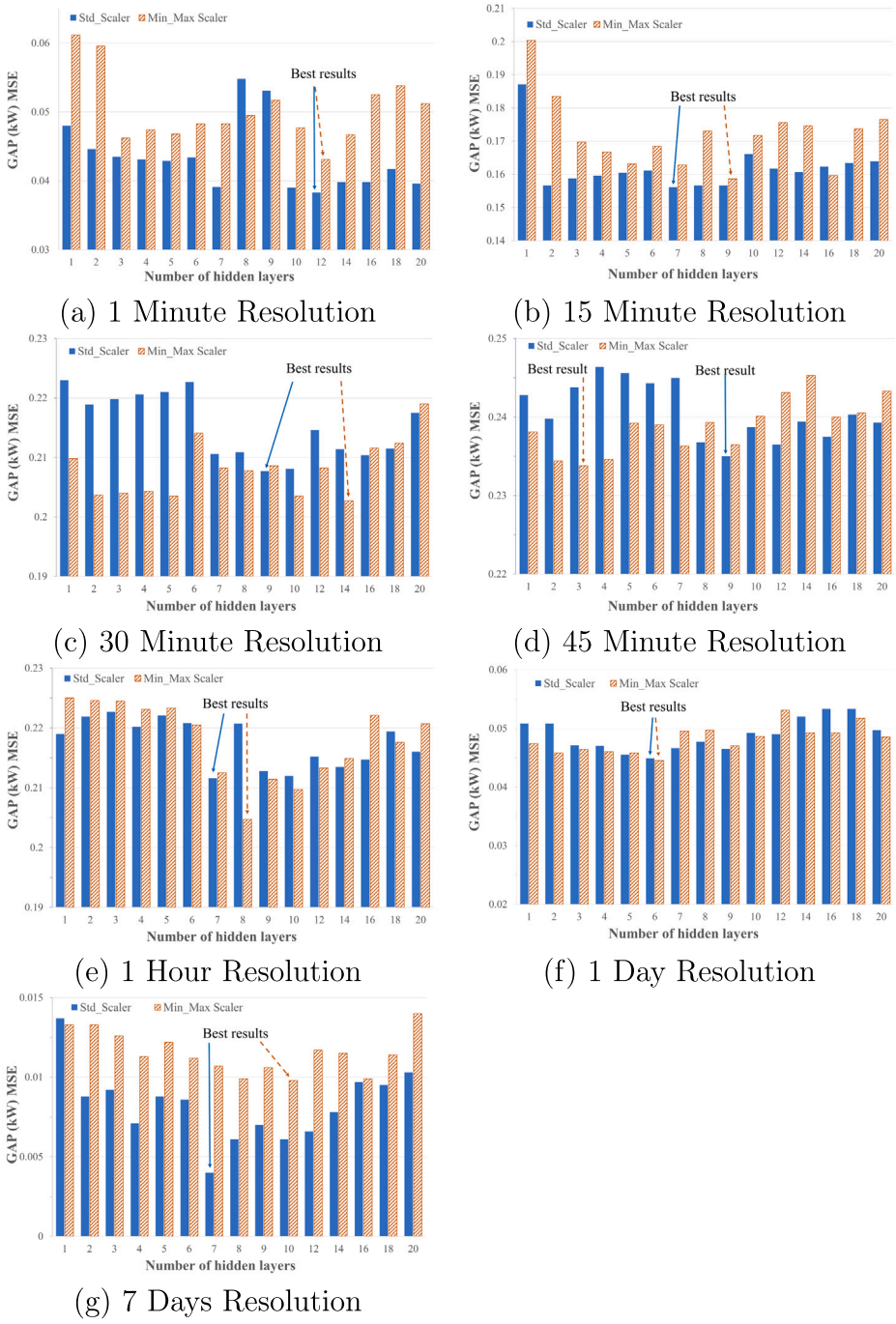


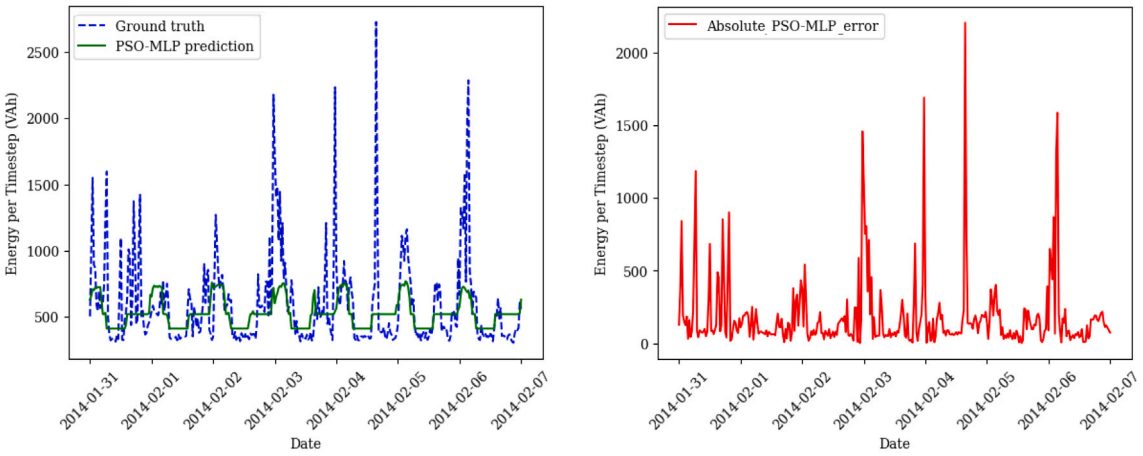
Fig. 5. UCI IHEPC PSO-MLP MSE for all architectures for both data standardisation techniques.

model [27]. The optimised MLP architecture in Table 4 is 29:53:58:62:66:52:44:34:19:9:1 (nine HL), with a learning rate of 0.01420 and a batch size of 189.

A FS 'threshold value' of 0.03 selected 12 from the original 29 features, and the resultant 'PSO-MLP FS' runs delivered improved performance across all three error metrics, as shown in Table 4. The optimal MLP architecture with FS is 12:72:67:82:72:58:45:23:14:1 (eight HL), with a learning rate of 0.00739 and a batch size of 77. These results demonstrate the clear benefit of hyperparameter optimisation combined with FS in achieving the best performance from an MLP. Fig. 6(a) displays the AMPds Whole House Energy profile for one week at the beginning of February 2014, labelled 'Ground truth', alongside the prediction from the PSO-MLP using FS. The PSO-MLP has learned an approximation of the daily energy profile, but it is unable to predict the larger intermittent loads

Table 4
AMPds - WHL measurement only: results comparison with advanced DL models from [27].

Error metric	FNN with DWT [27]	LSTM-DWT-CRT [27]	PSO-MLP	PSO-MLP FS
MAPE	47.13	44.44	36.12	33.22
MAE	283.70	276.44	266.08	254.14
MSE	166224	173077	157428	151214



(a) PSO-MLP prediction vs Ground Truth (b) PSO-MLP Absolute Error.

Fig. 6. AMPds (WHL) 1 Week resolution, PSO-MLP prediction vs Ground Truth and Absolute error.

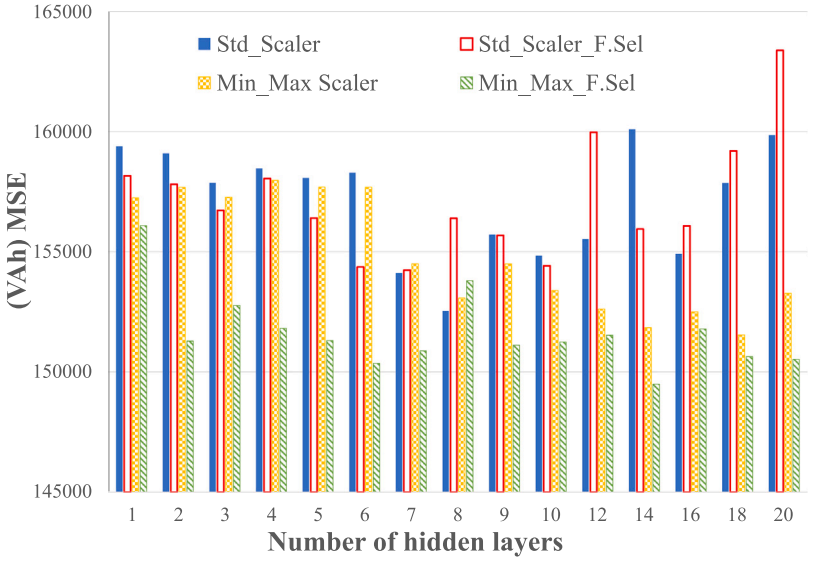


Fig. 7. AMPds (Six Loads) standardisation and FS results.

or the short-term transient load fluctuations. This limitation is not unexpected, as the model has no access to weather or occupancy data, nor to training data from the same time of year as the test data. Fig. 6(b) presents the absolute error for the same week, which is dominated by the large intermittent loads that the PSO-MLP finds difficult to predict accurately.

Two forms of data standardisation are employed, both with and without FS, with the MSE results displayed in Fig. 7 for the full range of MLP HL configurations searched for the AMPds dataset with six appliance measurements. The data for WHL follows a similar trend and is therefore not included. MSE is the error metric reported in Fig. 7. Whilst the performance across all models is similar with a single HL, the benefit of using FS with the Min_Max Scaler becomes evident from two HL onwards. Although the best result is observed at 14 HL, the MSE profile from two to twenty HL remains relatively consistent. In contrast, the Min_Max Scaler without FS shows gradual improvement with increasing HL, such that by eighteen HL, there is little added benefit from applying

FS. A different profile is observed when using the Standard Scaler, with the non-FS model peaking in performance at eight HL and then degrading as more HL are added. Fig. 7 highlights the importance of selecting an appropriate standardisation technique for the dataset, as well as the value of applying FS to improve predictive accuracy, particularly for shallower DL models.

4.3. Discussion

Improving building energy consumption prediction plays a key role in energy conservation [5] and enabling the control of smart grids [2] to optimise the use of local renewable power generation. Accurate STLF facilitates effective load shifting [6], helps reduce peak demand, and can lower building operational costs [7]. To achieve this, DL techniques can be employed to inform STLF by learning complex patterns in historical energy use, enabling predictive models — such as the proposed PSO-MLP — to anticipate future consumption and support real-time building energy management with a mean inference time of less than 3 ms. The prediction performance of the proposed PSO-MLP with the CESI N3 Smart Building case study data is illustrated in Fig. 2. The model successfully learned the daily cycles in the CESI data and significantly reduced the MAPE compared to manually set hyperparameters, achieving a 25.4% error reduction. This highlights the uplift in performance resulting from the proposed technique. However, it was less effective at predicting deviations from typical patterns. For example, the peak demand observed on Monday, 27 May was not well predicted. The prediction performance may be further improved by considering a wider range of inputs from the extensive CESI dataset (e.g. occupancy and/or HVAC control data) and exploring variations in the data processing.

This study set out to develop an approach that produces more accurate STLF predictions than advanced state-of-the-art DL techniques. Fig. 3 shows that the proposed PSO-MLP model has returned a lower MSE rate with the UCI IHEPC dataset at each time step compared to the foundational MLP and the MHA-CNN-LSTM model [26]. The relative improvement in performance grows with the time step, with the best result at 7-day resolution, and this trend is consistent throughout the range of results. Details within the input data are muted as the time resolution increases [12], and the proposed PSO-MLP model can better deal with this than the foundational MLP and the MHA-CNN-LSTM. One possible explanation is that the DL models proposed in [26] used one fixed set of hyperparameters for all time resolutions. In contrast, the proposed PSO-MLP approach automatically optimises the MLP hyperparameters separately for each run, which considerably improves the complexity of optimising DL architectures.

The PSO offers a practical advantage over exhaustive or grid-based search methods for hyperparameter tuning, particularly in high-dimensional spaces where the search landscape is non-convex and contains multiple local minima [78]. Existing search methods have exponential time complexity relative to the number of hyperparameter combinations; for a hyperparameter space with dimensions d and k discretised values per dimension, the total search cost grows as $\mathcal{O}(k^d)$ [79]. In contrast, PSO reduces this to approximately $\mathcal{O}(n \cdot g)$, where n is the number of particles and g the number of generations, typically with $n \ll k^d$, thus offering a substantial reduction in computational cost. For example, with $k = 20$ and $d = 4$, an exhaustive search requires 160,000 evaluations, whereas PSO with $n = 15$ and $g = 100$ requires only 1500. Building on this, the incremental optimisation strategy proposed in this study, where the number of HL is increased sequentially, further improves efficiency and convergence stability. In the implemented method, PSO operates within a 3-dimensional search space defined by the number of neurons per HL, learning rate, and batch size. For example, if the cardinality for each dimension is $|N|$, $|LR|$, and $|B|$, respectively, a total search space size will be $S = |N| \cdot |LR| \cdot |B|$ per HL setting. In contrast, processing HL as a fourth PSO dimension increases the search space to $S' = |N| \cdot |LR| \cdot |B| \cdot |HL|$, where $|HL|$ is the number of possible HL configurations. In addition, the incremental method avoids the complexity of mixed-type parameter optimisation, where HL (a structural parameter) interacts poorly with scalar hyperparameters in standard PSO update equations [80]. This decomposition improves the alignment between the swarm dynamics and the underlying optimisation landscape. As illustrated in Fig. 1 (Step 3), this structured search process ensures that deeper MLP configurations are only explored when earlier models indicate sufficient performance gains, while preserving control over the fixed time-constrained optimisation budget.

Fig. 5 illustrates the PSO-MLP performance using the UCI IHEPC data across all seven time resolutions. For all time resolutions, the best results are obtained using between three and fourteen HL, with a significant improvement evident when moving from one to more HL. This demonstrates the importance of exploring deep MLP architectures and the danger of assuming that a single-HL MLP can adequately model all scenarios. This finding aligns with that reported recently in [61]. Across the different datasets, we observed optimal MLP HL depths varying from 3, 9, 10 and 14. The literature [25] indicates that adding more HLs, up to a certain point, enhances the model's ability to extract complex nonlinear features from datasets. This includes interactions between HVAC cycles, weather and occupancy profiles, including across the lagged features. Beyond the optimal depth the risk of overfitting increases and training speed reduces. The variation in optimal depth across our datasets probably reflects the level of complexity in these nonlinear interactions between the features. Some required more layers (9–14) due to their high complexity, and others less (e.g. 3 layers) due to their simpler interactions. Additionally Fig. 5 reveals that at 15 and 45-minute and 1-day time resolutions, the two data standardisation techniques differ little in performance. However, the performance differential is significant for others, notably 7-day resolution, with the Standard Scaler performing best. This suggests that exploring different data standardisation techniques also has the potential to unlock additional prediction performance and should not be overlooked.

The results from the AMPds displayed in Fig. 7 support the same conclusions. The optimal performance is not found with a single HL but improves significantly when additional HL are added. The novel technique developed allows for iterative building and tuning of optimal hyperparameters. Rather than searching for an optimal solution for a deep MLP model without prior knowledge, it is consistently found to be more accurate to build upon previous optimal shallower solutions. The proposed PSO-MLP could build MLP models with between 60% and 25% lower error rates than the benchmark FFN with CRT or DWT and the advanced state-of-the-art DL LSTM-DWT-CRT could achieve.

Furthermore, FS is employed with the AMPds, and this enables the PSO-MLP to widen the performance gap to the comparator benchmark FFN and the state-of-the-art LSTM-DWT-CRT model [27] across all three error metrics, as seen in Tables 3 and 4. Fig. 7 reveals that FS improved the prediction performance when using the Min_Max Scaler, but was less consistently beneficial with the Standard Scaled data. However, it is important to note that FS was run with different threshold values for the two forms of data standardisation. Initially, both were run with a threshold of 0.01, and this gave an improvement for the Standard Scaler but not for the Min_Max Scaler. Consequently, the latter was rerun with a reduced threshold of 0.005, thus increasing the number of selected features from 19 to 61, and this resulted in the performance improvement evident in Fig. 7. The threshold used for the Standard Scaler may, therefore, be suboptimal, and indeed, a better threshold value might exist for the Min_Max Scaler too. Hence, there is scope to optimise the FS technique used in future studies. The Min_Max Scaler consistently returned better results for the AMPds than the Standard Scaler, again demonstrating the importance of assessing different standardisation options.

This study has demonstrated the effectiveness of the novel incrementally deepening PSO-MLP technique for electrical STLF (24 h ahead) in a small university building and two domestic houses, one in Canada and the other in France. The technique is expected to be generalisable to a wide range of buildings, particularly as its effectiveness has been demonstrated in smaller buildings with more erratic energy profiles. For larger buildings, the aggregation of individual electric load profiles produces a smoother cumulative load profile, which may further improve prediction accuracy [12].

5. Conclusion

This paper presents a novel incremental deepening PSO-based optimisation technique for MLP architectures that simultaneously tunes architectural and learning hyperparameters to enhance short-term building energy forecasting. The approach introduces an iterative strategy that progressively builds deeper networks from previously optimal shallow configurations, combined with integrated feature selection, a methodology not commonly explored in the STLF domain. This addresses key challenges in building energy management, as the energy supply sector remains the largest contributor to greenhouse gas emissions [1], making accurate energy consumption prediction critical for smart building operations and grid control.

The key findings demonstrate substantial improvements over baseline and advanced models across three datasets: 25.4% MAPE reduction compared to standard MLP in the CESI N3 Smart Building case study [16], 29.6% average improvement over MHA-CNN-LSTM [26] across UCI IHEPC dataset resolutions, and 15.32% performance gain over LSTM-DWT-CRT [27] on the AMPds dataset. The optimal architectures identified ranged from 3 to 14 hidden layers, with training times maintained under 130 min and mean inference times less than 3 ms, demonstrating computational efficiency alongside accuracy improvements. This enables more effective energy management by improving HVAC system responsiveness [7], supporting demand response strategies [6], and facilitating smart grid integration [2].

The practical implications for building energy management are significant, as the PSO-MLP model's lightweight computational requirements and efficient training make it suitable for real-time deployment in building management systems. Unlike resource-intensive hybrid deep learning models that present implementation challenges [64] and require GPU acceleration with substantial computational costs [66], the proposed approach enables HVAC optimisation, demand response strategies, and smart grid integration on standard BMS hardware, supporting more accessible smart building implementations and operational cost reductions. The technique is expected to be generalisable to larger buildings where load profile aggregation produces smoother patterns [12].

5.1. Limitations

The study acknowledges several limitations: the absence of systematic ablation analysis to isolate individual contributions of PSO optimisation, feature selection, and incremental depth; evaluation limited to relatively small buildings; exclusion of potentially valuable features such as occupancy and detailed HVAC data from the CESI dataset; and suboptimal feature selection thresholds that require further optimisation.

5.2. Future work

Future work should address these limitations through comprehensive ablation studies, validation on larger building types, systematic comparison with lightweight predictive alternatives (e.g. XGBoost, LightGBM and Transformer), and practical deployment studies focusing on BMS integration and digital twin implementation. Practical deployment requires further investigation of computing resource requirements, inference latency on standard BMS hardware, and optimal model update frequencies [14,15]. Field trials in operational buildings will be essential to validate these deployment parameters. Additionally, comparisons with existing network structure search methods, such as Neural Architecture Search, will be explored. The novel incremental swarm optimisation technique has broader applications in hyperparameter tuning for foundational and hybrid ML and DL methods across various practical areas [81], supporting interdisciplinary collaboration between AI methods and building engineering practice.

CRediT authorship contribution statement

Jess Robert Tindall: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kay Rogage:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Omar Doukari:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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